INCH-POUND
MIL-DTL-16377/38B(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/38A(SH)
9 May 1983

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; DETAIL LIGHTING FOR HULL NUMERALS, 120 VOLTS, WATERTIGHT, SYMBOL 304

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 5 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

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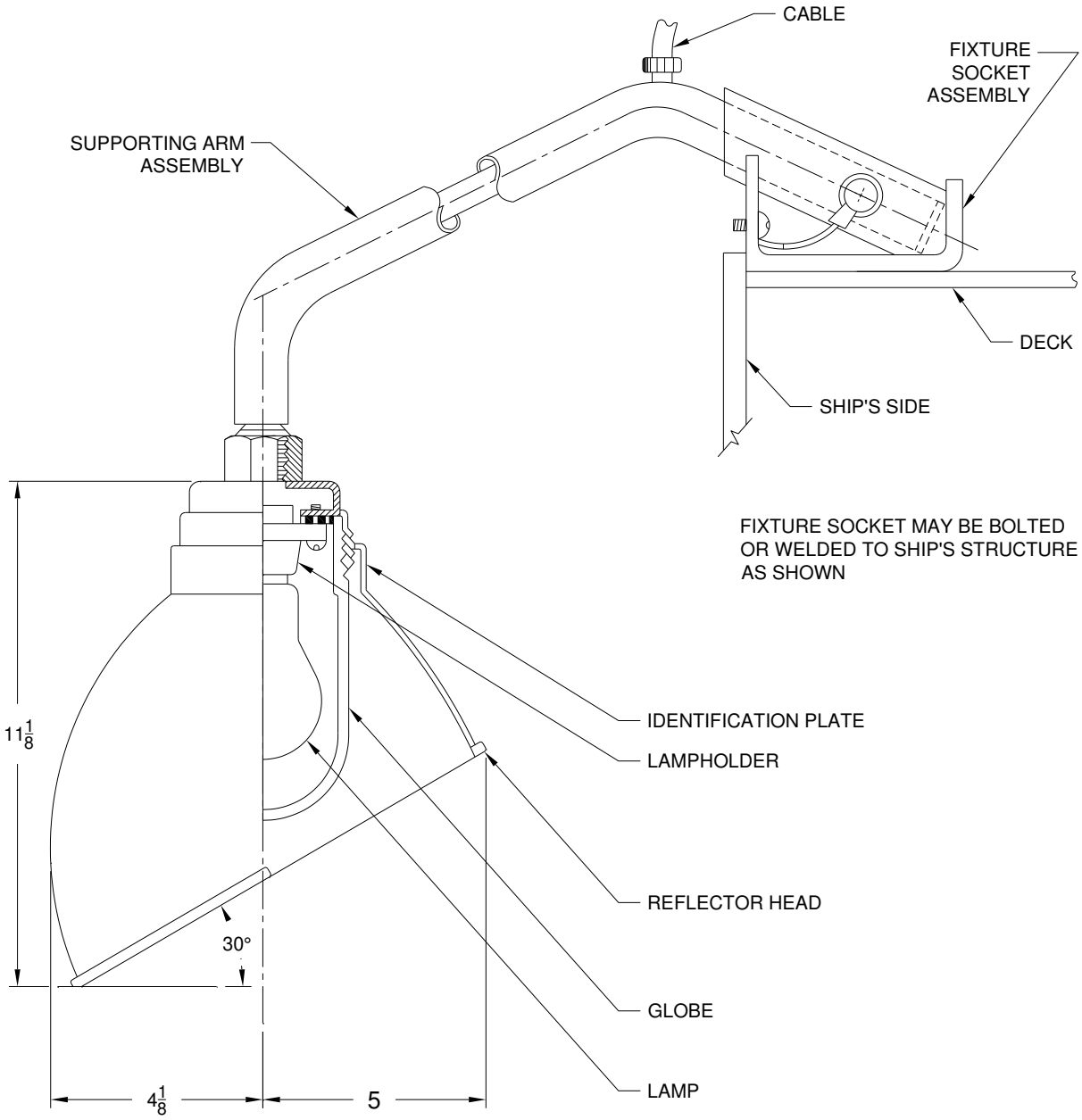
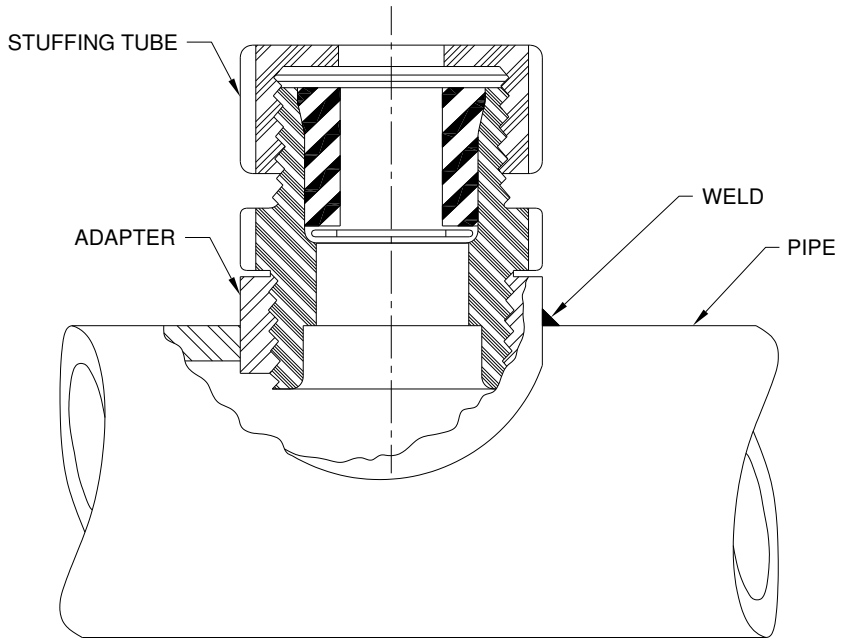
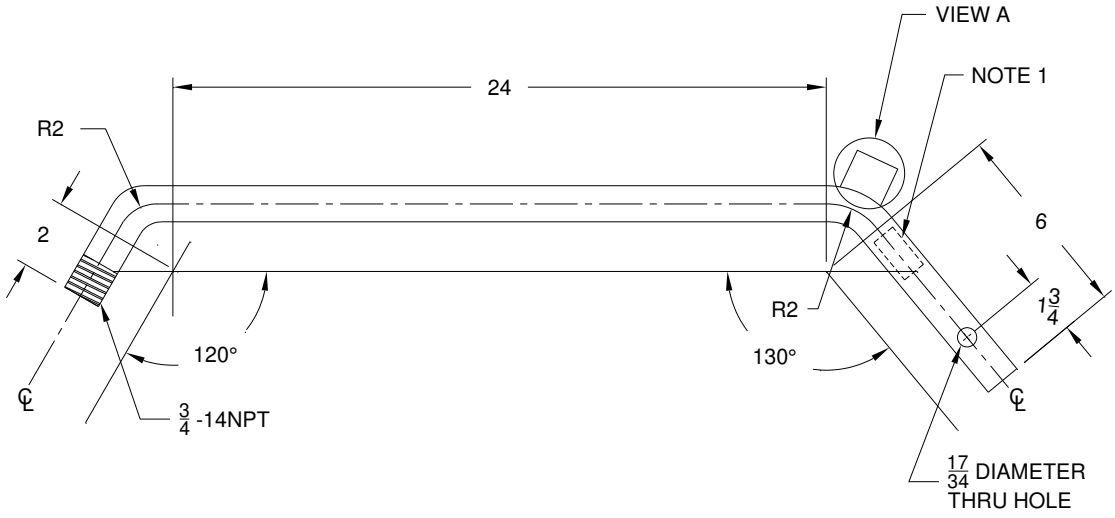


FIGURE 1. Typical view of symbol 304.

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VIEW A

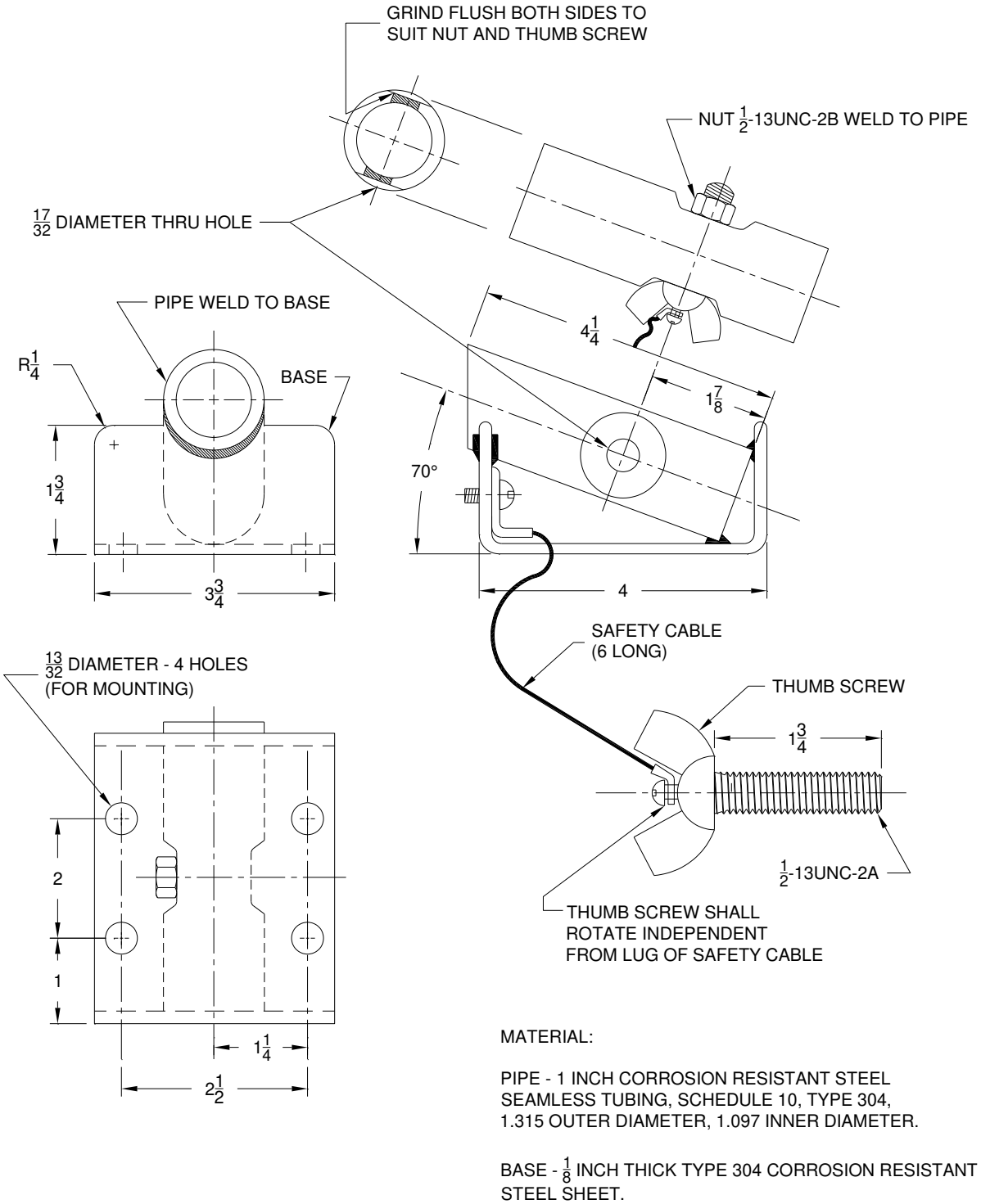


NOTES:

1. 4 inches from end press and cement-in a rubber stopper to seal tube end – watertight.
2. Material for pipe shall be ¾ inch corrosion-resistant steel seamless tubing, schedule 40, type 304, 1.050 od, 0.937 id.

FIGURE 2. Supporting arm assembly.

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MATERIAL:

PIPE - 1 INCH CORROSION RESISTANT STEEL SEAMLESS TUBING, SCHEDULE 10, TYPE 304, 1.315 OUTER DIAMETER, 1.097 INNER DIAMETER.

BASE - 1/8 INCH THICK TYPE 304 CORROSION RESISTANT STEEL SHEET.

FIGURE 3. Fixture socket assembly.

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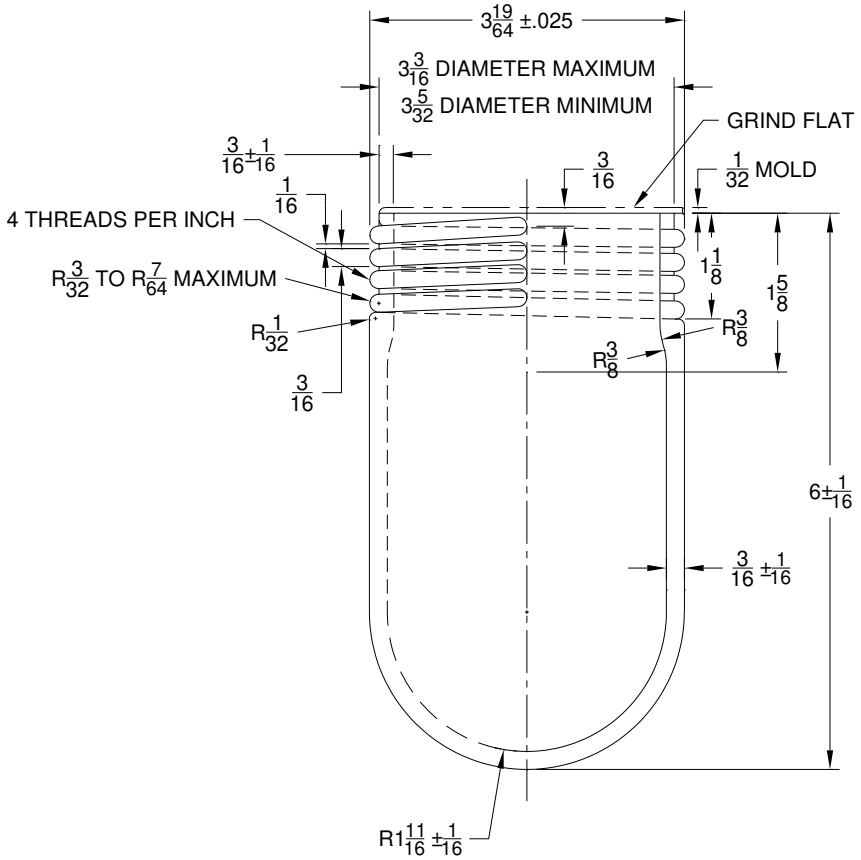


FIGURE 4. Globe.

MIL-DTL-16377/38, SYMBOL 304

MFR'S NAME_____

NOTE: Material shall be corrosion-resistant steel, 0.020 inch thick. Letters shall be ⅛ inch high, filled with black enamel.

FIGURE 5. Identification plate.

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REQUIREMENTS

CLASSIFICATION

Type: II or III
Class: 2

SYMBOL NUMBER

304: Light for lighting bow and hull numerals for use with one 120-volt lamp.

DESIGN

Electrical rating:	100 watts (incandescent) or 50 watts (solid-state lighting [SSL]), 120 volts.
Dimensions and configuration:	See figures 1 and 2.
Supporting arm and fixture socket:	Galvanized steel. A sealing compound shall be used for watertightness at installation of supporting arm to connection box.
Reflector head and connection box:	Commercial type for sign illumination.
Safety cable:	See figure 3. A 1/32-inch inner diameter stainless steel cable with a 1/16-inch outer diameter black vinyl covering. Terminal shall be copper or copper alloy in accordance with MIL-T-16366, type WTG.
Effectiveness of enclosure:	Watertight.
Lampholder:	UL 496, Supplement SB, medium screw, molded rubber.
Cable:	10 feet; MIL-DTL-915/8, type TSS-3.
Electrical connections:	Electrical connections shall be made watertight.
Grounding lead:	Red grounding lead shall be connected to lampholder securing screw and to appropriate connection in the plug or outlet box.
Stuffing tube:	ASTM F1836M, size 2.
Packing assembly:	ASTM F1836M, size 17.
Lamp:	This fixture shall be designed to accommodate a plastic-coated 100-watt, 120-volt incandescent lamp, industry number 100A21/RS (plastic coated) or an SSL lamp with 1,200 lumen minimum output. (Lamp not furnished.)
Finish:	The inside surface of the reflector head shall have a white, baked enamel finish. Exterior surfaces shall be finished in accordance with MIL-DTL-16377.
Identification plate:	See figure 5.

EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table I.
Conformance inspection:	
Comparison inspection:	Same as the first article inspection.
Conformance inspection:	See MIL-DTL-16377.

GENERAL INFORMATION

Military part number: M16377/38-304.

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TABLE I. First article and comparison inspection.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Enclosure effectiveness	3.5.11	4.8.14	Watertight 1/
Shock	3.5.12	4.8.7	Grade B
Vibration	3.5.13	4.8.8	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Salt fog	3.3.15	4.8.10	---
Grounding circuit	3.6.13.3	4.8.11	---
Magnetic permeability	3.4.8	4.8.18	---
Continuity of grounding	3.6.13.4	4.8.12	---

NOTE:
1/ Electrical circuit only.

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6210-2023-015)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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